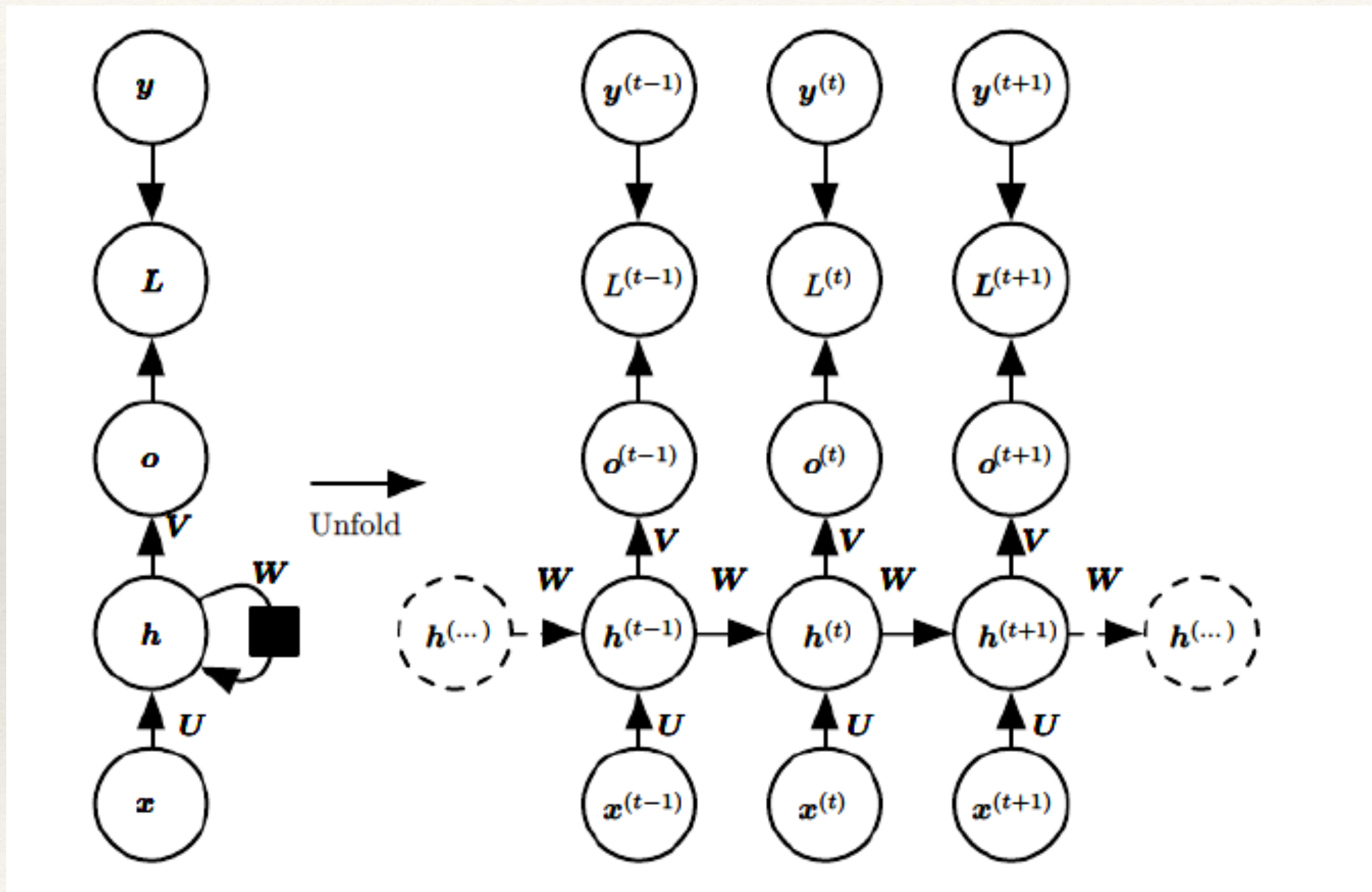


E9 205 Machine Learning for Signal Processing

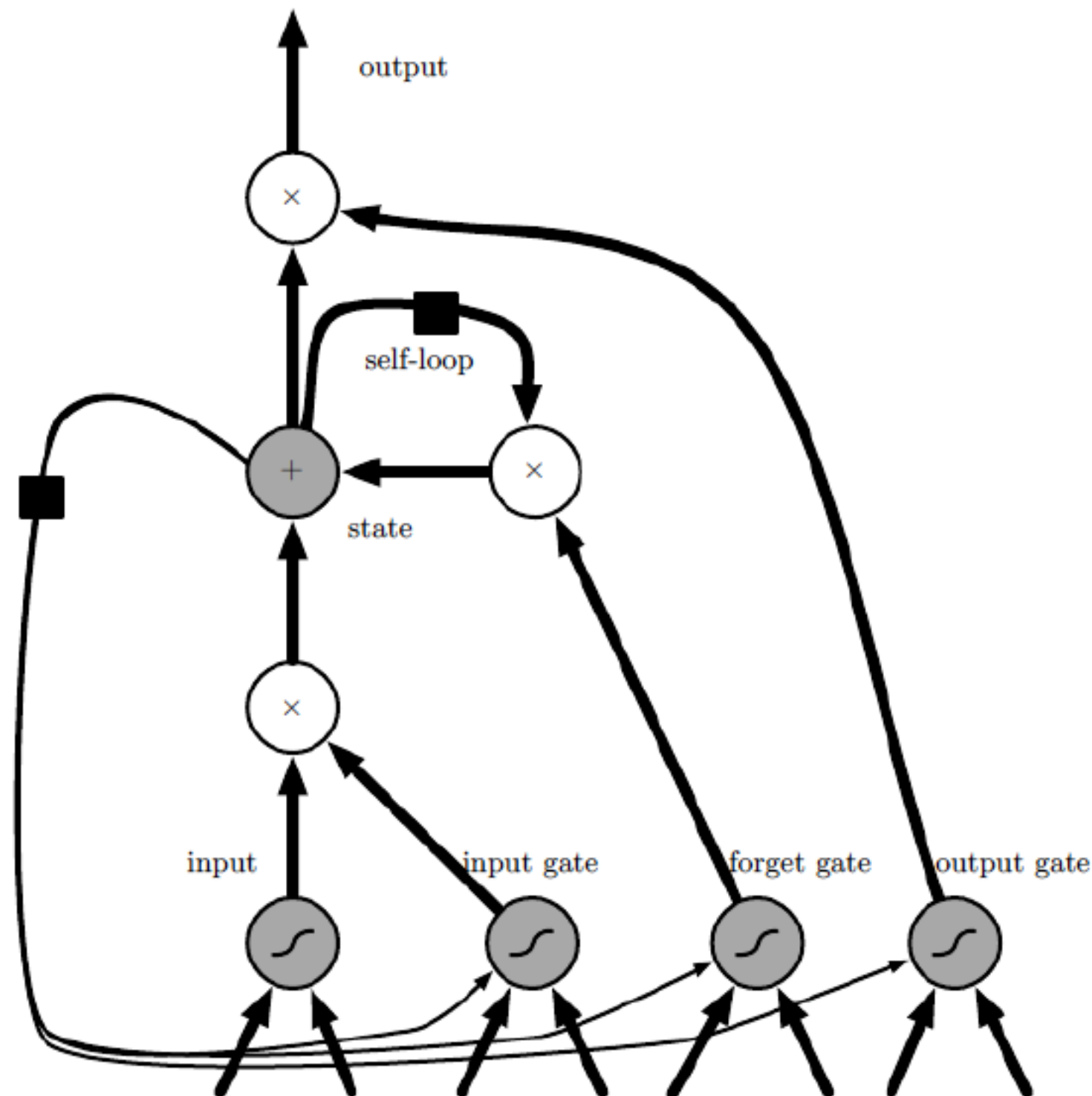
Convolutional and Recurrent Networks

18-11-2017

Recurrent Networks



Long Short Term Memory Networks



Long Short Term Memory Networks

$$f_i^{(t)} = \sigma \left(b_i^f + \sum_j U_{i,j}^f x_j^{(t)} + \sum_j W_{i,j}^f h_j^{(t-1)} \right)$$

$$s_i^{(t)} = f_i^{(t)} s_i^{(t-1)} + g_i^{(t)} \sigma \left(b_i + \sum_j U_{i,j} x_j^{(t)} + \sum_j W_{i,j} h_j^{(t-1)} \right)$$

$$g_i^{(t)} = \sigma \left(b_i^g + \sum_j U_{i,j}^g x_j^{(t)} + \sum_j W_{i,j}^g h_j^{(t-1)} \right)$$

$$h_i^{(t)} = \tanh \left(s_i^{(t)} \right) q_i^{(t)}$$

$$q_i^{(t)} = \sigma \left(b_i^o + \sum_j U_{i,j}^o x_j^{(t)} + \sum_j W_{i,j}^o h_j^{(t-1)} \right)$$

Restricted Boltzmann Machines

A Symmetrical, Bipartite, Bidirectional Graph with Shared Weights

